OIML Certification System	CMI	V 3112

OIML Member State	**OIML Certificate No.**
Czech Republic	R76/2006-A-CZ1-26.02

OIML CERTIFICATE ISSUED UNDER SCHEME A

OIML Issuing Authority

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Manufacturer

Name: TENZOVÁHY, s.r.o.
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Identification of the certified type *(the detailed characteristics will be defined in the additional pages)*

Non-automatic weighing instrument, single range
type TENZOWIM 134

Designation of the module *(if applicable)*

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This OIML Certificate attests the conformity of the above identified type (represented by the sample(s) identified in the OIML type evaluation report) with the requirements of the following Recommendation of the International Organization of Legal Metrology (OIML):

OIML R 76 Edition (year): 2006

For accuracy class: III

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This OIML Certificate relates only to metrological and technical characteristics of the type of measuring instrument covered by the relevant OIML Recommendation identified above.

This OIML Certificate does not bestow any form of legal international approval.

The conformity was established by the results of tests and examinations provided in the associated reports:

Test report No. 6012-PT-R0003-26 dated 25.6.2026 that includes 52 pages.
Test report No. 8551-PT-E0231-18 dated 30.10.2018 that includes 1+55 (Test report + annex) pages.
Test report No. 8552-PT-S0012-26 dated 22.6.2026 that includes 7+5 (Test report + annex) pages.

OIML type evaluation report 0511-ER-W028-23 dated 25.6.2026 that includes 7 pages.

The technical documentation relating to the identified type is contained in documentation file:

0511-UL-W028-23

OIML Certificate History

Revision No.	Date	Description of the modification
-	21 July 2026	Issuing of certificate

The OIML Issuing Authority
RNDr. Pavel Klenovský
Director of Certification Body

Date: 21 July 2026

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Important note: Apart from the mention of the Certificate's reference number and the name of the OIML Member State in which the Certificate is issued, partial quotation of the Certificate and of the associated OIML type evaluation report(s) is not permitted, although either may be reproduced in full.

1 Instrument name and description

The weighing instrument TENZOWIM 134 is designed for determining the weight of wheels, axles, and the total weight (gross vehicle weight) of road vehicles. The weighing instrument operates in non-automatic mode with a single weighing range.

1.1 Weighing instrument modules

The weighing instrument consists of the following parts (modules):

- Load receptor(s) in a foundation frame
- Load cells
- WIMCONTROL indication unit with legally relevant software (LRSW): WSERVER, IOUNIT, WTERMINAL)

Optional auxiliary devices:

- Heating/cooling of measuring modules for operation in the extended temperature range
- Signalling devices: traffic lights, acoustic horn
- Clients of the WSERVER server application that are used, for example, for secondary data display and print, LRD post-processing according to the user's needs, etc.
- Printer
- External display
- Devices to control vehicle entry into the weigh zone (barriers, retractable pillars and similar traffic control devices)
- Number plate readers - video detection
- Devices for detecting vehicle presence and measuring vehicle dimensions (induction loops, RFID chip readers, optical barriers, cameras, radar detectors, etc.)

1.1. Principle of operation

The WIMCONTROL indication unit of the scales processes signals from the load cells located under the load receptor which is situated in between aprons (in the middle of weighing zone.
Low-level signals from the load cells, which varies accordingly with force caused by vehicle wheels acting on the load receptor, are amplified, converted into a digital signals and converted into mass units.

2 Weighing instrument design

2.1 Mechanical design

Two versions of the load receptor are possible, consisting of a single weighbridge constructed as a one-piece steel weldment or of two weighbridges that include two separate steel weldments as in the following figures:

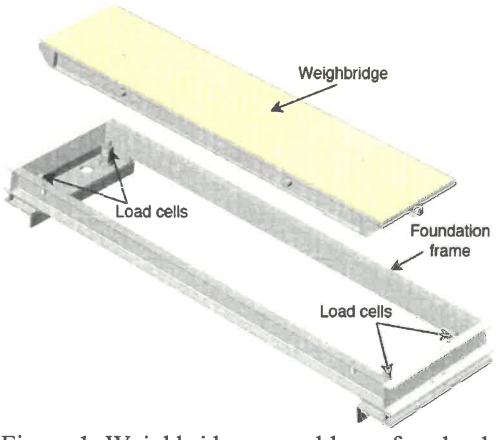


Figure 1: Weighbridge assembly on four load cells embedded in the foundation frame (single)

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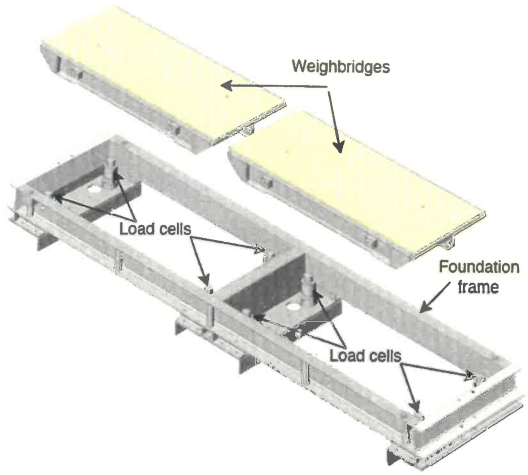


Figure 2: Two weighbridges each on four load cells embedded in the foundation frame (double)

Four load cells are located under the weighbridge.

The foundation frame of the weighbridge(s) is used to place the weighbridge at the level of the surrounding aprons.

2.2 Design of the weigh zone

Weigh zone comprise the load receptor in a frame in advance of and beyond each end of the load receptor in the direction of travel of the vehicle being weighed. Foundation frame with weighbridge(s) is levelled to a plane in both longitudinal and transversal directions.

2.3 Load cells

TENZOWIM 134 uses analogue compression transducers with OIML R 60 certificate of conformity, accuracy class C3 or higher.

2.4 WIMCONTROL indication unit

The indication unit of the weighing instrument consists of two CCPU and IOUNIT modules that communicate together using a UDP protocol via an Ethernet interface.
The modules may be integrated into a common COMPACT chassis as shown in Figure 3, or into standalone SPLIT chassis as shown in Figure 4.:

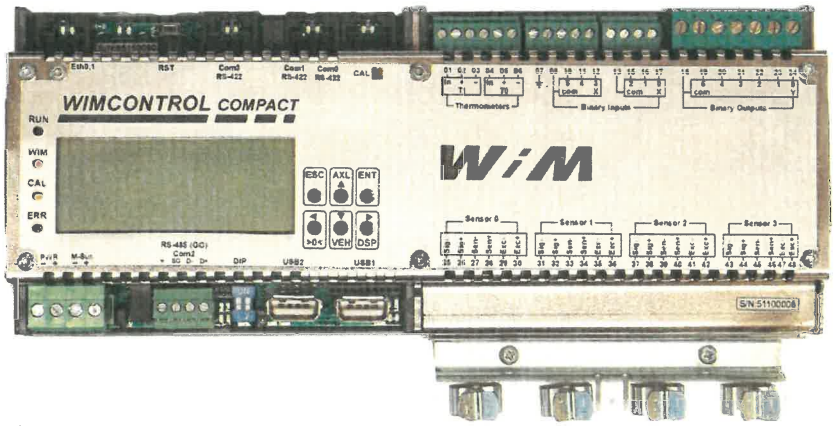


Figure 3: WIMCONTROL indicator unit – COMPACT version

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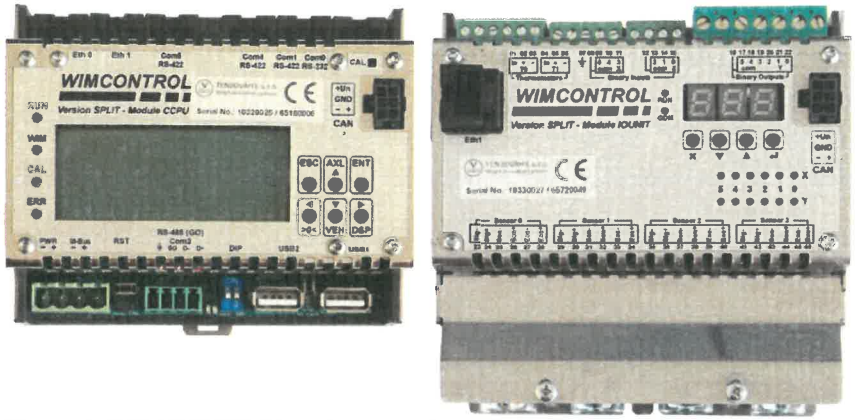


Figure 4: WIMCONTROL indicator unit - CCPU module and IOUNIT module in SPLIT version

In the production process, both modules of SPLIT version, which forms the WIMCONTROL indication unit, are paired together by a unique code that prevents later replacing of the IOUNIT or CCPU module without breaking the verification mark placed on the CAL switch on the CCPU module. In addition, the pairing process generates a new adjustment number that is displayed each time the unit is switched on.

2.4.1 IOUNIT input and output module

The IOUNIT module processes data samples coming from:

- 1 up to 4 analogue inputs for load cells,
- 1 internal and up to 2 external temperature sensors,
- Up to 6 binary inputs for auxiliary devices,
- Up to 6 binary outputs for auxiliary devices.

Analogue signals from load cells are converted in the IOUNIT module into digital signals and compiled together with digital signals into communication packet. The packet is secured by CRC code against modification and sent to the CCPU processor module for processing.
Data from temperature sensors are used to monitor the actual operating temperature of the measuring chain.

2.4.2 CCPU processor module

The CCPU processor module processes data from the IOUNIT modules and provide vehicle weighing. If multiple IOUNIT modules are communicating with CCPU module, the IOUNIT modules are connected via an Ethernet switch.
Each weighing result is stored in the internal integrated memory with a DSD function, which maintains legally relevant data obtained from weighing.
The CCPU module has an alphanumeric display for primary indication of measured values. External software clients of the WSERVER server application may be used for secondary indication or post-processing of data.

2.4.3 Switchboard design, temperature measurement & control

The WIMCONTROL indication unit, equipped with either one COMPACT module or two SPLIT modules, is housed in an EMC-shielded enclosure called the "control unit," which meets the IP 43 standard for protection against external influences and the electromagnetic compatibility (EMC) requirements for measuring instruments. The Control Unit contains filter circuits for WIMCONTROL and features connectors for connecting load cells, temperature sensors, binary inputs/outputs, power supply, and an Ethernet communication interface. The load cell connectors are secured against unauthorized tampering.
The control Unit is housed in a switchboard that ensures suitable operating conditions for the weighing modules and for electronics of optional devices connected to the weighing system.
If the electronic components are installed in an environment with extreme temperatures exceeding the standard range of operating temperatures, the switchboard must be double-shell design with forced air circulation between the walls.

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The switchboard further includes:

a. A thermostat-regulated heating unit. The heating unit maintains the temperature inside the switchboard within the standard range of operating temperatures even if the ambient temperature drops to the lower limit of the extended range of operating temperatures.
b. The switchboard box may optionally be fitted with an air conditioner. The air conditioner maintains the temperature inside the switchboard within the standard range of operating temperatures even if the ambient temperature rises to the upper limit of the extended range of operating temperatures.

If the temperature inside the switchboard or within the load receptor with the load cells exceeds the operating temperature limit despite these measures, the WIMCONTROL unit displays an alert and prevents further weighing.

3 Software

The legally relevant software of the scales and its security meet the requirements of the WELMEC 7.2:2023 software guide.
Following software applications running on the hardware of the WIMCONTROL indication unit are legally relevant (hereinafter referred as "LRSW"):

1. The Linux operating system - Debian distribution with LinuxRT (Embedded),
2. WSERVER - legally relevant software that provides the basic functions of the scales for vehicle weighing and external communication of the WIMCONTROL unit,
3. WTERMINAL - legally relevant software used for operating the keypad and display (a human interface device - "HID") of the CCPU module of the WIMCONTROL indication unit,
4. IOUNIT FIRMWARE - firmware of the IOUNIT module of the indication unit, which processes signals from the load cells, thermos sensors and binary inputs/outputs into digital form and transmits them to the CCPU module for evaluation.

Applications 1 to 3 are stored in the CCPU module's secured internal integrated memory and application 4 is stored in the IOUNIT module's secured internal integrated memory.
The clients can connect to the indication unit and the internal DSD memory only via the LRSW WSERVER. The clients does not contain any legally relevant functions and cannot influence legally relevant functions or change data in the DSD memory.

3.1 Software modules and their functions

3.1.1 WSERVER

At the client's request, WSERVER weighs, evaluates and stores legally relevant data into the integrated DSD memory.
The legally relevant functions of the LRSW WSERVER are as follows:

- Saving LRF parameters acquired during scales calibration and verification
- Static weighing
- Weighing-in-motion
- Saving weighing records in the internal integrated DSD memory
- User account administration - sign in, sign out, user account creation, modification and deletion
- Transmitting LRD (weighing data stored in the DSD memory) via an Ethernet interface

3.1.2 WTERMINAL

The WTERMINAL is the LRSW operating the HID devices - the alphanumeric display and the keypad on the CCPU module.
The following legally relevant functions may be operated using the WTERMINAL application:

- User account administration
- Logging user activities and displaying the logs
- Static weighing
- Weighing-in-motion

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- Displaying/printing/exporting the weighing records stored in the DSD memory
- Displaying the LRSW (versions and checksums) of the WTERMINAL, WSERVER and FIRMWARE IOUNIT applications
- Changing the WTERMINAL menu language
- Configuring the network parameters of the Ethernet interface
- Setting date and time (manually or through synchronization via an ntp server)
- Exporting weighing records from the DSD memory into an external storage medium via a USB interface

When the WIMCONTROL indication unit is switched on and the LRSW WTERMINAL starts, the user BOOT is automatically signed on, indicating only the current status of the scales in static mode.
The remaining functions are available only for users logged in with higher-level username and password.

3.1.3 IOUNIT firmware

The firmware of the IOUNIT module is LRSW stored in the integrated memory of the IOUNIT module. The firmware processes input signals from the IOUNIT module inputs (3 inputs for thermometers, 6 binary inputs, 4 inputs for load cells) and controls the status of six binary outputs. The firmware processes the acquired data, logs the data and transfers it to the WSERVER application, which runs in the paired CCPU module, for further processing.
Data transmission between the IOUNIT and the CCPU takes place via an Ethernet interface, using a UDP protocol. Depending on the unit version, an internal bus (COMPACT version) or an Ethernet 1 interface (SPLIT version) is used for communication.
The IOUNIT FIRMWARE does not have any user interface or any adjusting features. The IP address, gateway and mask required for communication with the CCPU module are set up during module production process and cannot be changed by the user.
Communication between the CCPU and the IOUNIT takes place in a separate network segment that is dedicated solely for the mutual communication of these modules.

3.2 Software identification by version number and CRC

LRSW applications are identified by version number in following manner:

- WSERVER: WSRV-1.01.02
- WTERMINAL: WTRM-1.01.02
- IOUNIT Firmware: IOUN-1.01.02

The display of LRSW version numbers are included in the indicator's start-up sequence. Scrolling down, the Adjustment No. (see below) and CRC codes of the LRSW applications are displayed.
These data are also accessible during system operation from the indicator user menu.

3.3 Adjustment number mechanism

To prevent unauthorised changes to LRF parameters, the system is further protected by an adjustment number mechanism. Changes to LRF parameters always result in a new adjustment number generated by the LRSW WSERVER application, which is displayed upon start-up of the WIMCONTROL unit:

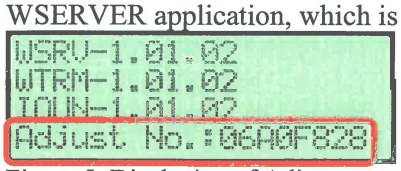


Figure 5: Displaying of Adjustment number

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3.4 Storage and displaying of legally relevant data from weighing

LRD generated during the weighing process are automatically stored in the integrated DSD memory of the indication unit upon completion of weighing. External access to the DSD memory (excluding access controlled by the LRSW WSERVER application) is disabled.
The data in the DSD memory is secured against intentional and unintentional changes by a checksum. After each write to the DSD memory, the checksum value is recalculated and stored in the LRSW WSERVER configuration.
Quantity of stored weighing records is set to 1 000 000 by default (can be changed in factory parameters).
Viewing records in the DSD memory is possible by client's query to WSERVER. WSERVER reply with data dump from the DSD to display of the Indication unit or to communication interface.

3.5 Service client WSERVICE and adjusting of LRF parameters

The WSERVICE service client provides functions for servicing and calibration of the scales, and adjusting their LRF parameters. Writing LRF parameters to the DSD memory is performed exclusively by the WSERVER.
Changes to LRF parameters can be applied only when 'CAL' switch is activated on the WIMCONTOL unit.

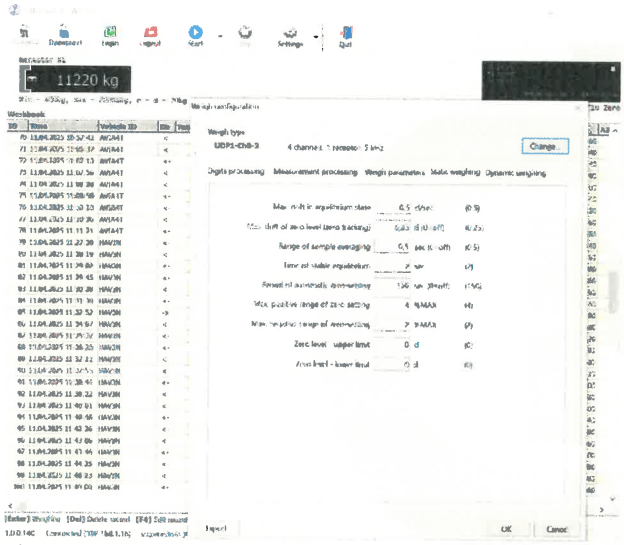


Figure 6: WSERVICE client

4 Technical data and metrological characteristics

4.1 Weighbridges technical data

Weighing platforms (weighbridges) are manufactured in two types - "Single" and "Double", dimension of each type can be adjusted to customer requirement in the following range:

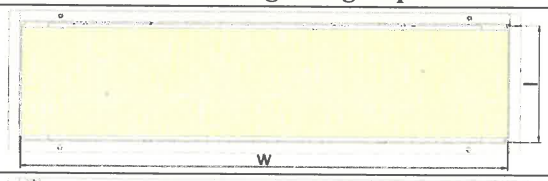
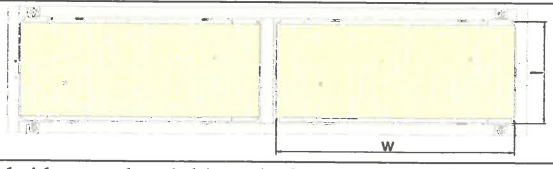
Weighbridge type	Weighbridge top view	Active surface [mm]
Single T20 **Single T30**		Length (l): 500 - 900 Width (w): 2800 - 4000
Double T20 **Double T30**		Length (l): 500 - 900 Width (w): 1200 - 2000

Table 1: Types of weighbridges and weighing platform(s) dimensions

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The ***Length*** of the platform refers to the dimension measured in the direction of vehicle ride. The ***Width*** of the platform refers to dimension measured across the direction of vehicle ride.

Dimensional variability of load receptors do not affect the metrological characteristics or LRF of the scales.

4.2 Metrological characteristics

Weighbridge type	**Single T20**		**Single T30**		**Double T20**		**Double T30**	
Accuracy class	(III)							
Max [kg]	20 000		30 000		10 000		15 000	
Min [kg]	200	400	200	400	200	400	200	400
d = e [kg]	10	20	10	20	10	20	10	20
n	2000	1000	3000	1500	1000	500	1500	750
Temperature range standard	- 20 to + 50°C							
Temperature range extended	- 40 to + 70°C							

Table 2: TENZOWIM 134 metrological characteristics

4.2.1 Wheel load, Axle load and Total vehicle weight

With the *Single* type of weighbridge, Min and Max, as shown in Table 2, determine the range of weight capacity *per axle of the vehicle* (Min_{axle}, Max_{axle}).
The weighing range for *the total vehicle weight* (Gross vehicle weight - GVW) is determined as the product of the weighing range of one axle and the maximum number of weighable axles (A_{max}).

$$Max_{GVW} = (2 \times Max_{wheel}) * A_{max}$$

In the default configuration of TENZOWIM 134 scales, the parameter $A_{max} = 16$. The value of A_{max} can be adjusted during the manufacturing process in the range $16 \leq A_{max} \leq 99$.

With the *Double* type of weighbridge, Min and Max, as shown in Table 2, determine the range of weight capacity *per wheel of the vehicle* (Min_{wheel}, Max_{wheel}).
For the weighing capacity range *of one axle,* the following formula applies:

$$Min_{axle} = 2 \times Min_{wheel}$$
$$Max_{axle} = 2 \times Max_{wheel}$$

The weighing range for *the total vehicle weight* (Gross vehicle weight - GVW) is determined as the product of the weighing range of one axle and the maximum number of weighable axles (A_{max}).

$$Max_{GVW} = (2 \times Max_{wheel}) * A_{max}$$

4.3 Power supply

Weighing instrument is powered by 230V AC / 50 Hz.

5 Interfaces and optional peripheral devices

TENZOWIM 134 uses the Ethernet interface as the primary communication channel. The USB interface is a secondary channel which is used exclusively for the dump of DSD memory with weighing records to USB flash drive. Both communication interfaces are exclusively controlled by LRSW WSERVER.
Communication over other hardware interfaces is prohibited on the system level of Linux RT OS.

TENZOWIM 134 weighing device can be equipped with optional auxiliary devices providing additional functions of the weighing device. These devices are connected to and controlled by external SW client running on an external device, typically on a PC with Windows OS or a tablet with Android OS.
Peripheral devices and its data do not affect the LRD of the weighing instrument.

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5.1 Printing equipment

The printing device ensures the printing of weighing tickets. The printed output contains an identifier enabling validation of the weighing record with LRD in the DSD memory of the indication unit.

5.2 External display

The external display provides the display of vehicle weighing data and other information about the operation of the scales.

5.3 Vehicle presence identification devices

Depending on the method of use of the scales, the scales can be equipped with an auxiliary device(s) for vehicle identification. Such devices include: RFID chip readers; photoelectric barriers; camera and ANPR systems; radar detectors (and others).

5.4 Signalling devices

Various types of visual or acoustic signalling devices, such as traffic lights, directional arrows, barriers, horn, retractable pillars etc., allow the scale operator to control vehicle traffic over the scale.

5.5 Vehicle recognition devices

Depending on the purpose of its use, the weighing instrument may be equipped with an auxiliary vehicle recognition device that starts weighing once the presence of a vehicle is detected. Vehicle recognition devices are, for example: RFID chip readers, inductive loops, optical barriers, ultrasonic sensors, camera systems (for video detection or for licence plate reading) and radar detectors.

6 Verification and securing of the weighing instrument

The metrological tests must be carried out according to national applicable regulations.

6.1 WIMCONTROL indication unit and its securing

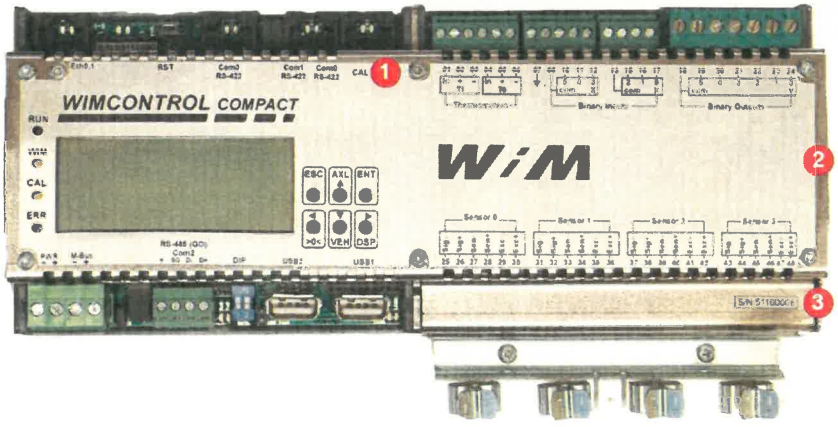


Figure 7: Placement of verification marks on WIMCONTROL COMPACT version

Following areas of WIMCONTROL COMPACT indication unit are secured with verification marks:

(1) CAL switch
(2) Interface between the top and side cover of the unit
(3) Cover over connectors from load cells

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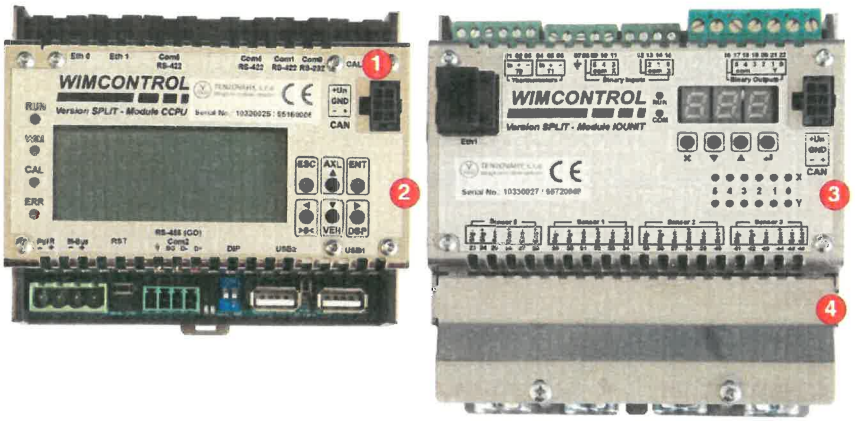


Figure 8: Placement of verification marks on the WIMCONTROL SPLIT version

Following areas of WIMCONTROL SPLIT indication unit are secured with verification marks:

(1) CAL switch on CCPU module
(2) CCPU module - interface between the top and side cover of the unit
(3) IOUNIT module - interface between the top and side cover of the unit
(4) IOUNIT module - cover over connectors from load cells

6.2 Main instrument label

The main instrument label with complete descriptive markings is located on the weighing instrument's indication unit. The label complies with clause 7.1 of the international recommendation OIML R 76. Following information are provided on the label:

TENZO
CE M YY
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Instrument type Weighbridge type	TENZOWIM 134 Single T20	OIML R 76 class	III	e [kg]	20	Adjustment No.	02B1F816
Serial number	K0310001	OIML R 134 class	---	Oper. speed [km/h]	---	WSRV version	1.01.02
OIML R 76 certificate No.	TCM 128/YY- XXXX	Min [kg]	400	Temp. range [°C]	-20 ÷ +50	WTRM version	1.01.02
OIML R 134 certificate No.	---	Max [kg]	20 000	Power supply	230V AC / 50 Hz	IOUN version	1.01.02
Producer	• TENZOVÁHY, s.r.o. • Mechatova 345/3 • 783 01 Olomouc • • Czech Republic • www.tenzovahy.com •					Manufactured	2024

Figure 9: Main instrument label of instrument with "Single" type of weighbridge

TENZO
CE M YY
XXXX

Instrument type Weighbridge type	TENZOWIM 134 Double T30	OIML R 76 class	III	e [kg]	10	Adjustment No.	03B2C212
Serial number	K0320001	OIML R 134 class	---	Oper. speed [km/h]	---	WSRV version	1.01.02
OIML R 76 certificate No.	TCM 128/YY- XXXX	Min wheel / axle [kg]	200 / 400	Temp. range [°C]	-20 ÷ +50	WTRM version	1.01.02
OIML R 134 certificate No.	---	Max wheel / axle [kg]	15 000 / 30 000	Power supply	230V AC / 50 Hz	IOUN version	1.01.02
Producer	• TENZOVÁHY, s.r.o. • Mechatova 345/3 • 783 01 Olomouc • • Czech Republic • www.tenzovahy.com •					Manufactured	2024

Figure 10: Main instrument label with "Double" type of weighbridge

The label is secured with a verification mark.

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